

Nishan Dananjaya

📍 Kurunegala, Maho ✉ eg19118@seu.ac.lk ☎ +94 718834578 in Nishan Dananjaya
🏠 W.D.N.D. Bandara k Nishan Bandara

Profile

A dedicated Electrical and Electronics Engineering undergraduate with a passion for continuous learning and innovation. Committed to staying at the forefront of emerging technologies, problem-solving, and applying engineering principles to design, develop, and implement cutting-edge solutions. Driven by a strong desire to enhance technical skills and contribute meaningfully to projects in diverse, challenging environments.

Education

South Eastern University of Srilanka

2021 - Present

B.Sc. (Hons.) Electrical and Electronics Engineering

- GPA: 3.78/4.0
- **Coursework:** Power systems, Electrical machines and drives, Electrical installations, Electromagnetics, Digital and analog electronics, Power electronics, Signal and systems, Communications theory and systems, Computer architecture

Experience

Trainee Electrical and Electronic Engineer

BEPZ, Malwana

North Manufacturing PVT. LTD.

Oct 2023 - Jan 2024

- Designed a computer vision model to detect defects in tapes used for sail manufacturing, improving quality control.
- Developed a humidity control room for yarn storage to ensure optimal conditions.
- Designed a motor control circuit using the (DOL) starter method for the humidity control room.
- Engineered various types of molds for sail production, optimizing efficiency and product durability.

Projects

Heart Disease prediction using ML

github.com/Heart Disease Prediction 

- Used different types of algorithms such as linear regressions, Neural network, XG boost and etc
- Tools and platforms: Jupyter notebook, Tensorflow, Scikitlearn, Streamlit, Python

Automatic defects detecting system using YOLO V5

github.com/defect detection YOLOV5 

- Developed a system to detect the defects of the tapes that has been used to make sails
- Updated the system to save the defected tape images with annotations
- Tools Used: YOLO V5, YOLO V8, Pytorch, Python

Gemini Chatbot Sinhala and Tamil

github.com/Gemini chatbot sinhala 

- Using the Gemini API built a chat-bot that can respond in Sinhala and Tamil language. it also capable of giving audio responses.
- Tools Used: Gemini pro API, Streamlit, Python, Deep translator from google and Google text to speech.

MCQ generator

github.com/MCQ Generator 

- This application allows users to generate multiple-choice questions (MCQs) from a given PDF. Users can upload a PDF and specify parameters such as the complexity level and the desired number of MCQs. The app then generates MCQs along with the correct answers.
- Tools Used: Gemini pro API, Streamlit, Python, Pypdf, Langchain

Programmable Gyroscope sensor using STM32

github.com/Gyro_Sensor_PCB 

- PCB design using the MPU 6050 sensor and STM32 microcontroller for motion sensing. Integrated power management and connectivity features for robotics and IoT applications.
- Tools Used: Altium

LIPO battery charger

github.com/Lipo_battery_charger 

- Designed a LIPO battery charger using charge controller
- Tools Used: Altium

Real-time driver drowsiness detection system

github.com/drowsiness_testing 

- Developing a computer vision model for real-time detection of driver drowsiness, integrated with an alert system to notify the driver to take a rest, enhancing road safety and reducing fatigue-related accidents.
- In the future, we aim to design a hardware accelerator to enhance the performance and efficiency of this system, enabling faster real-time processing and improved responsiveness.
- Tools Used: Tensorflow, Python, Kaggle platform to train the model

Technologies

Programmng Languages: Python, C++, JavaScript

ML and DL: Tensorflow, Pytorch, Scikit learn, Pandas, Numpy, Open cv, Seaborn

Front end: HTML, CSS

Technologies: Linux shell

Software and tools : Altium, Matlab, Solidworks, LT spice, Proteus

Extra Curricular activities

IEEE student branch

2022 - present

Member, Organizer

SEDS agni (Robotics and rovers division)

2022 - 2023

Member

Reference

Prof. M.A.L.A. Haleem (malahaleem@seu.ac.lk)

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Dr. Mahinda Alahakoon (mahinda@seu.ac.lk)

Former Senior Lecturer and Head,
Department of Electrical and Telecommunication Engineering,
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South Eastern University of Srilanka,
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